

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0783

Northern States Flambeau 100 kV
September 26, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0783. The Interconnection Customer proposes a 145.7 MW solar photovoltaic facility interconnecting at the Northern States Flambeau 100 kV substation in the Ohio area. The request was received on April 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 309 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$59,428,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Franchise Tax – Northern States Flambeau 100.0 kV Circuit 1: 142.4% loading (156 MVA) following the outage of the Franchise Tax Substation – Northern States Flambeau 100.0 kV Line 2; DFAX = 27.5%.

The Franchise Tax – Northern States Flambeau 100.0 kV Line is overloaded to 142.4% of its 156 MVA emergency rating for the loss of the Northern States Flambeau – Franchise Tax 100.0 kV Ckt 1. The project contributes 27.5% DFAX (40.0 MW) to this overload.

The Filer 500 kV/345 kV TX is overloaded to 126.6% of its 745 MVA emergency rating for the loss of the Lerdo Substation – Tidd 500.0 kV. The project contributes 5.5% DFAX (8.1 MW) to this overload.

For the loss of W Coshocton – West Astoria 100.0 kV L, the Los Vaqueros – Colleyville 100.0 kV loads to 121.9% of its 138 MVA rating. Project DFAX: 11.7%.

Colleyville – Franchise Tax 100.0 kV: 119.8% loading (149 MVA) following the outage of the Colleyville Substation – Franchise Tax 100.0 kV Cir. 2; DFAX = 17.0%.

The Franchise Tax – Colleyville 100.0 kV Ckt 2 is overloaded to 119.8% of its 149 MVA emergency rating for the loss of the Franchise Tax – Colleyville 100.0 kV Cir. 1. The project contributes 17.0% DFAX (24.8 MW) to this overload.

The Los Vaqueros – Tewksbury No. 22A 100.0 kV Line is overloaded to 112.0% of its 137 MVA emergency rating for the loss of the Trail – Colleyville Substation 100.0 kV. The project contributes 10.6% DFAX (15.4 MW) to this overload.

Stevens Rd – Conesville Station 230.0 kV: 105.1% loading (516 MVA) following the outage of the Conesville Station – Mukingum Riv 500.0 kV; DFAX = 6.9%.

Atascocita – Maple Street 230.0 kV: 104.5% loading (563 MVA) following the outage of the El Paso – Ravenna Substation 500.0 kV; DFAX = 6.0%.

For the loss of Hilltop – Piney Crk Project 500.0 kV, the Ritter Park – Sugar Hill 345.0 kV line loads to 102.7% of its 1,886 MVA rating. Project DFAX: 10.3%.

For the loss of Lions Rd – Piney Crk Project 500.0 kV, the Ganso – El Paso 500.0 kV loads to 101.9% of its 2,779 MVA rating. Project DFAX: 13.1%.

For the loss of Trail – Colleyville 100.0 kV, the Benson Power Biomass Plant Substation – Tewksbury No. 22A 100.0 kV loads to 100.9% of its 132 MVA rating. Project DFAX: 10.6%.

Handsome Lake – Piney Crk Project 500.0 kV: 100.4% loading (2,649 MVA) following the outage of the Lions Rd – Piney Crk Project 500.0 kV line; DFAX = 11.6%.

For the loss of Ganso – El Paso 500.0 kV, the Lions Rd – Piney Crk Project 500.0 kV loads to 100.0% of its 3,166 MVA rating. Project DFAX: 12.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Northern States Flambeau – Francise Tax 100.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$13,625,000	100% (\$13,625,000)
Franchise Tax – Northern States Flambeau 100.0 kV Line	Rebuild line with higher-capacity conductor	\$7,192,000	100% (\$7,192,000)
Filer 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$28,284,000	4% (\$1,283,000)
Los Vaqueros – Colleyville 100.0 kV	Rebuild line with higher-capacity conductor	\$5,108,000	100% (\$5,108,000)
Franchise Tax – Colleyville Substation 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$10,754,000	100% (\$10,754,000)
Franchise Tax – Colleyville 100.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$12,458,000	100% (\$12,458,000)
Los Vaqueros – Tewksbury No. 22A 100.0 kV	Rebuild line with higher-capacity conductor	\$3,941,000	100% (\$3,941,000)
Stevens Rd – Conesville Station 230.0 kV	Reconductor line	\$12,187,000	4% (\$464,000)
Maple St – Atascocita 230.0 kV line	Reconductor line	\$11,655,000	6% (\$730,000)
Sugar Hill – Ritter Park 345.0 kV Line	Reconductor line	\$6,836,000	2% (\$127,000)
Ganso – El Paso 500.0 kV	Reconductor line	\$55,973,000	2% (\$1,038,000)
Benson Power Biomass Plant – Tewksbury No. 22A 100.0 kV	Reconductor line	\$1,112,000	100% (\$1,112,000)
Handsome Lake – Piney Crk Project 500.0 kV	Reconductor line	\$35,171,000	2% (\$675,000)
Lions Rd – Piney Crk Project 500.0 kV	Reconductor line	\$50,196,000	2% (\$921,000)
	Total allocated to project		\$59,428,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.